

# A Method of Distance Measurement by Using Monocular Camera

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**Abstract:** In this paper, we propose the method to measure the distance to the surface with any figures. The method uses two images, one is taken at a camera point and the other is taken at the point where the camera is moved along the optical axis. The distance is calculated from the ratio between each size of an object projected on two images. From the fact that the pair of images has concentric circles features, the ratio is obtained by using Complex-Log Mapping. This method can calculate the distance to the surface with texture feature or not.

## 1. Introduction

In recently, many methods to measure the distance between the surface of an object and a camera have been developed. Stereo vision is the most popular method because of using no special light. But, if there is texture feature on the surface, the distance is not measured by using this method because it is impossible to search corresponding point of two images. When the texture feature is known beforehand, monocular vision is able to measure the distance[1], however, if the feature is unknown, the distance can not be done by any method. Therefore, it is necessary to develop the method which is able to measure the distance to any figured surface. So, in this paper, we propose a new method to measure the distance to the surface which has any figure by using monocular camera.

In the proposed method, we use two images, one is taken at a camera point and the other is taken at the point where the camera is moved along the optical axis of a camera. The distance between the surface and the camera is measured by the ratio which is calculated from two projected sizes of same object on these images. As the considering image is of the surface with unknown texture, it is impossible to measure the ratio directly. So, instead of measuring

directly we use a feature that one image is obtained by radially magnifying or reducing the other image. And the center of changing is the intersection point between the optical axis of a camera and the image plane, namely, the center of image. From the feature, let us consider the correspondence between two images on the perpendicular coordinates representation that the origin of the coordinates is the center of image. Each corresponding points exist in the same direction when observing from the origin. And the ratio between two distances to the origin matches to the ratio between two images. The feature is called concentric circles feature. So, to obtain the ratio of two images, we adopt Complex-Log Mapping (CLM)[2] that is a kind of polar coordinates transformations. Because, in the transformation, when a pair of images has the concentric circles feature, two mapped images have interesting feature. If the ratio between mapping radii corresponds to the ratio of two images, two mapped images become same image. By using the feature of CLM, we developed the new method to calculate the ratio of two images.

## 2. Principle of proposed method

The principle of the proposed method is shown in Fig. 1. Where  $a$  is the size of a

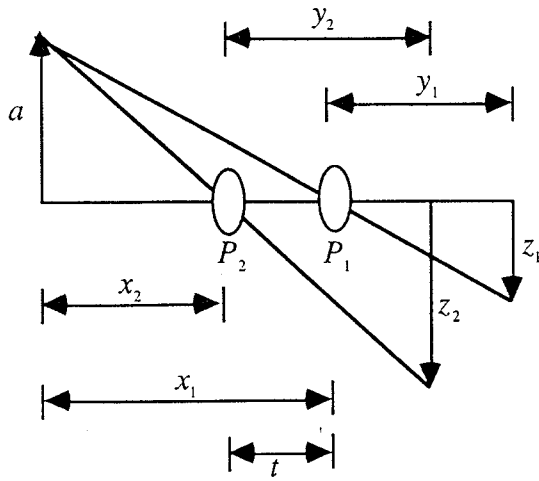


Fig. 1 Principle of measurement

considering object,  $x$  is the distance between the object and a camera,  $y$  is the distance between the camera and a projected plane (image),  $z$  is the size of the object projected on the image and  $f$  is a focal length of the camera. Then, these equations (1) and (2) are obtained from lens formula.

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{f} \quad (1)$$

$$z \cdot x = a \cdot y \quad (2)$$

If the camera is moved by the decided distance  $t$  from  $P_1$  to  $P_2$  along the optical axis, following equations (3) and (4) are obtained.

$$x_1 - kx_2 - f(1 - k) = 0 \quad (3)$$

$$t = |x_1 - x_2| \quad (4)$$

Where  $z_1$  is the size of the object on an image at position  $P_1$ ,  $z_2$  is the size at position  $P_2$  and  $k$  is the ratio of  $z_1$  to  $z_2$ . From now, the ratio of  $z_1$  to  $z_2$  is called the ratio between two images.

If  $k$  is obtained,  $x_1$  and  $x_2$  are calculated from following equations (5) and (6).

$$x_1 = f - \frac{k}{1 - k} \cdot t \quad (5)$$

$$x_2 = x_1 - t \quad (6)$$

## 2. Complex-Log Mapping

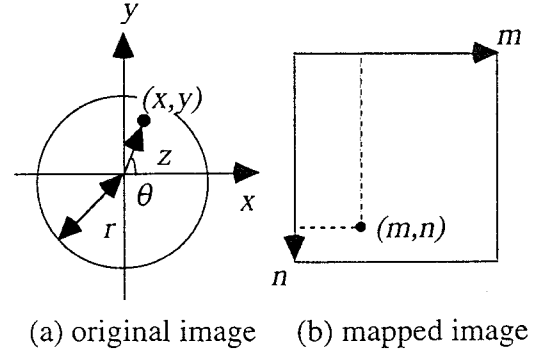


Fig. 2 Principle of CLM

Fig. 2 shows the principle of CLM. CLM transforms pixels in the area of the mapping radius  $r$  as centering a noticeable point. As shown in Fig. 2 (a), on the perpendicular coordinates centering it, component values of a pixel are  $(x, y)$  and the brightness value of the pixel is  $f(x, y)$ . In Fig. 2 (b), the brightness value  $f(x, y)$  on the original image is transformed to  $f_c(m, n)$  on the mapped image. Then, each correspondence is expressed in these equations (6).

$$f_c(m, n) = \frac{z}{r} \cdot f(x, y)$$

$$m = N \cdot \frac{\theta}{2\pi}, \quad n = N \cdot \log_r z \quad (6)$$

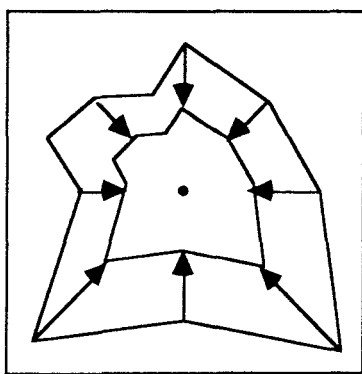
$$\theta = \tan^{-1} \frac{y}{x}, \quad z = \sqrt{x^2 + y^2}$$

Where  $N$  is the size of mapped image and plus direction of  $\theta$  is counterclockwise.

Each axis of  $m$  and  $n$  are related to the distance and the direction from mapping origin respectively.

## 3. Method of calculating the ratio

Now, suppose that two images are changed by two CLMs with same structure respectively



• : center of image

Fig. 3 Corresponding points between two images

when each image has concentric circles feature. The structure is that the mapping origin is the center of image and the mapping radius is same in two CLMs. The center of image is the intersection point between the optical image and image plane. As shown in Fig. 3, between two original images, the rotational movements of each corresponding point are not surveyed. Therefore, the component value of the axis  $m$  on each mapped image is equal but each that of the axis  $n$  is not. The ratio between each that of the axis  $n$  corresponds to the ratio between two images. And the component value of the axis  $n$  is obtained by the ratio between the radius and the distance from the origin on original image. Here, consider that each of original images is done by each CLM with different structure respectively. Between two CLMs, the origin is same position but the radius is different. So, if the ratio of two radii matches to it between two images, two mapped images by each CLM become same image from mentioned above. By using the feature of CLM, we developed the new method to calculate the ratio between two images.

This method uses two mapped images of CLM transformation. As the mapping origin

is the center of image, first mapped image is obtained by transforming one original image, which is called basic image from now on, in range of a mapping radius. Another mapped image is got from sequentially changing the other original image, which is called derivative image. In each CLM of derivative image, the mapping origin is unchanged. But the mapping radius is changed into it which is made to lengthen or shorten basic image's gradually. Then the correlation of two mapped images is consecutively calculated. The ratio of two radii that minimize the correlation is  $k$  in equation (5). Then, the correlation  $C$  is expressed in equation (7).

$$C = \frac{\sum_{j=1}^n \sum_{i=1}^m (I_1(i, j) - \mu_1)(I_2(i, j) - \mu_2)}{\sqrt{\sum_{j=1}^n \sum_{i=1}^m (I_1(i, j) - \mu_1)^2 \sum_{j=1}^n \sum_{i=1}^m (I_2(i, j) - \mu_2)^2}} \quad (7)$$

Where  $I_i(i, j)$  is the brightness of the coordinates values  $(i, j)$  on each mapped image,  $\mu_i$  is the average value.

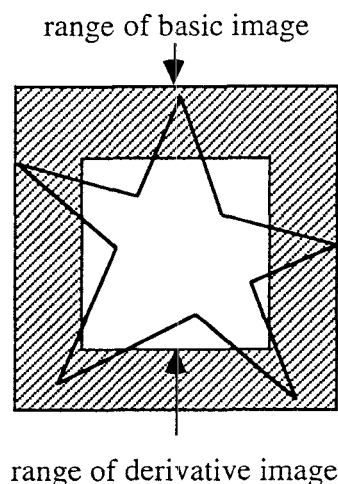


Fig. 4 Difference between basic image and derivative image

When derivative image is transformed, it is necessary to consider the effect of sampling

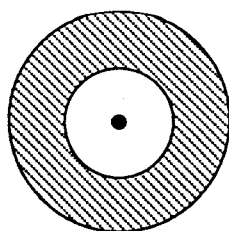
errors. So, CLMs are performed at each pixels around the center of image as the mapping origin.

Let us consider whether the mapping radius is lengthened or shortened when derivative image is changed. If the radius is lengthened, the distance from the surface which is taken as basic image is inevitably longer than that as derivative image. Therefore, the visual field of basic image is wider than that of derivative image. As the result, derivative image can not include all elements of basic image as shown by shaded portion in Fig. 5. And the portion causes the error to the calculating of the correlation between two mapped images. So, the portion must be excluded from the area of CLM. But, it is impossible to remove it because the surface has unknown texture feature. For that reason, the radius is shortened so that derivative image is able to contain all elements of basic image.

#### 4. Method of reducing noises, etc.

In proposed method, the ratio between two images is calculated from the pretended displacement of each corresponding point by moving a camera. For the reason, it is necessary to take care of the sampling error, noises and so on. So, in order to make one image, plural images are taken, added and averaged.

For escaping the sampling errors, in CLM, the brightness of a pixel is obtained by the average value within a small area centering it.



● : center of image

Fig. 5 Range of CLM

In this paper, a camera is installed on a robot arm. So, the axis of moving the camera almost coincides with the optical axis. Therefore, the displacements between corresponding points is restricted on the line which joins the center of image to each position. But around the center, the displacement is very small when the camera is moved. So, if only each projected size on two images is different by one pixel, the ratio changes too greatly. Moreover, the sampling error gravely influences the precise calculation of the ratio between two images. Therefore, CLM is executed only in the region except for the vicinity of the center. Then, as shown by shaded portion in Fig. 5, the range of CLM becomes shape like donuts.

#### 5. Experiment and the result

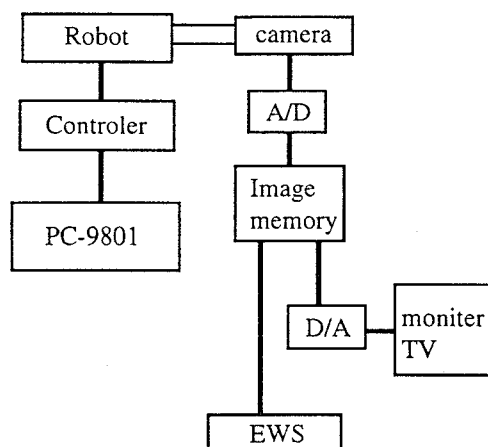


Fig. 6 Block diagram of experimental system

Fig. 6 is shown block diagram used in this research. The focal length of lens used in this research is 16.02mm.

The size of original image used in the experiment is 384 and the size of mapping image is 256. Then, the maximum of mapping radius is 128 and the minimum is 32. Some samples of images used as basic image in experiments are shown in Fig. 7, 8, 9 and 10. Fig. 7 is taken *Pressed cork* from, Fig. 8 is *Reptile skin* and Fig. 9 is *Grass lawn* from TEXTURE

PICTURES by P. Bordatz, respectively. And Fig. 10 is taken part of calendar. Fig. 7, 8 and 9 have texture features and Fig. 10 does not. The same method is used in each experiment and no information for the feature of image is given. Each Table 1, 2, 3 and 4 expresses each result of measurements applied to each image respectively. Then, there is only the result of the position where basic image is taken in each table.

From experimental results, it is proved that the proposed method is available to both surfaces which have texture feature and does not have it.

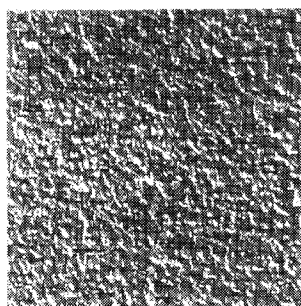


Fig. 7 *Pressed cork*

Table 1 Result of *Pressed cork*

| move-<br>ment<br>distance<br>of<br>camera<br>(mm) | Measure-<br>-ment<br>distance<br>(mm) | Real<br>distance<br>(mm) | Ratio $k$ |
|---------------------------------------------------|---------------------------------------|--------------------------|-----------|
| 50                                                | 499.6                                 | 500                      | 0.906     |
| 100                                               | 498.1                                 | 500                      | 0.828     |
| 150                                               | 506.3                                 | 500                      | 0.766     |

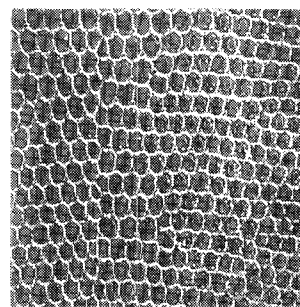


Fig.8 *Reptile skin*

Table 2 Result of *Reptile skin*

| move-<br>ment<br>distance<br>of<br>camera<br>(mm) | Measure-<br>-ment<br>distance<br>(mm) | Real<br>distance<br>(mm) | Ratio $k$ |
|---------------------------------------------------|---------------------------------------|--------------------------|-----------|
| 50                                                | 499.6                                 | 500                      | 0.906     |
| 100                                               | 498.1                                 | 500                      | 0.828     |
| 150                                               | 506.3                                 | 500                      | 0.766     |

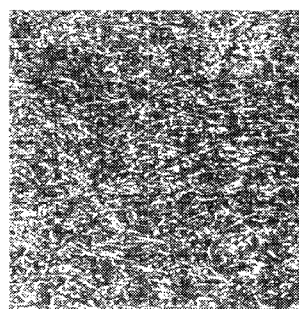


Fig. 9 *Grass lawn*

Table3 Result of *Grass lawn*

| move-<br>ment<br>distance<br>of<br>camera<br>(mm) | Measure-<br>-ment<br>distance<br>(mm) | Real<br>distance<br>(mm) | Ratio $k$ |
|---------------------------------------------------|---------------------------------------|--------------------------|-----------|
| 50                                                | 499.6                                 | 500                      | 0.906     |
| 100                                               | 498.1                                 | 500                      | 0.828     |
| 150                                               | 506.3                                 | 500                      | 0.766     |

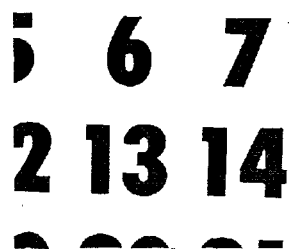


Fig. 10 part of calendar

Table 4 Result of calendar

| move-<br>ment<br>distance<br>of<br>camera<br>(mm) | Measure-<br>-ment<br>distance<br>(mm) | Real<br>distance<br>(mm) | Ratio $k$ |
|---------------------------------------------------|---------------------------------------|--------------------------|-----------|
| 100                                               | 498.1                                 | 500                      | 0.828     |
| 150                                               | 506.3                                 | 500                      | 0.766     |

## 6. Conclusions

In this paper, we proposed a new method for calculating the distance to texture surface from the ratio between two images taken by monocular camera moved on the optical axis. When the ratio was obtained, we used Complex-Log Mapping from the correspondence between two images which has concentric circles features. From the experiment result, the validity of this method was shown.

In this research, when calculating the ratio of two images, the sequential method is used. As a result, the processing time is long. Therefore, we would like to try different method to calculate the ratio. And the object surface of measurement is restricted in the plane which vertically intersects the optical axis. So, In the future research, we'll want to adopt this method for arbitrary plane.

## Reference

- [1] A. P. Witkin, " Recovering surface shape and orientation from texture", Artificial Intelligence, Vol. 17, pp17-45, 1981
- [2] K. Daniilidis, "Attentive Visual Motion Processing: Computations in the Log-Polar Plane", Computing, Springer-Verlag, pp 1- 20, 1996.